

ESP32-Based Environmental Monitoring System using DHT22 and LDR

1. Introduction

In this project, an environmental monitoring system is designed using the ESP32 microcontroller. The system is capable of measuring temperature, humidity, and ambient light intensity using sensors. With the increasing importance of smart monitoring systems in real-world applications, this project demonstrates how basic sensors can be interfaced with a microcontroller to collect and display environmental data in real time.

The implementation was carried out using the Wokwi simulation platform, which allows testing without physical hardware.

2. Objectives

- To interface a DHT22 sensor with ESP32 for temperature and humidity measurement
- To measure light intensity using a photoresistor (LDR)
- To read analog and digital sensor data using ESP32
- To display sensor readings on the serial monitor
- To simulate the system using Wokwi

3. Components Used

- ESP32 Microcontroller
- DHT22 Temperature and Humidity Sensor
- LDR (Light Dependent Resistor) Module
- 10k Ω Resistor (Pull-up for DHT22)
- Connecting Wires
- Wokwi Simulation Platform

4. System Description

The system consists of two main sensors:

- **DHT22 Sensor:** Measures temperature and humidity and sends digital data to the ESP32.
- **LDR Sensor:** Detects light intensity. The analog output varies depending on the amount of light falling on it.

The ESP32 reads data from both sensors and processes it. The results are then displayed on the serial monitor for observation.

5. Circuit Connections

- **DHT22:**
 - VCC → 3.3V
 - GND → GND
 - DATA → GPIO 4
 - 10kΩ between VCC & DATA
- **LDR:**
 - VCC → 3.3V
 - GND → GND
 - AO → GPIO 34

6. Working Principle

The DHT22 sensor continuously measures temperature and humidity and sends the data digitally to the ESP32. The microcontroller reads this data using a dedicated library.

The LDR module produces an analog voltage depending on light intensity. When the surroundings are bright, the resistance decreases, resulting in a higher analog value. In darker conditions, the value decreases.

The ESP32 reads both values and prints them on the serial monitor. Based on the LDR value, the system categorizes the environment as "Bright" or "Dark".

7. Code Explanation

- The DHT library is used to read temperature and humidity values.
- GPIO 4 is configured for DHT22 data communication.
- GPIO 34 is used as an analog input pin for LDR.
- The `analogRead()` function reads light intensity.
- The `Serial.print()` function displays all sensor values.

CODE:

```
#include <DHT.h>

#define DHTPIN 4    // DHT22 DATA -> ESP 4

#define DHTTYPE DHT22

#define LDR_PIN 34  // Analog pin for LDR (AO -> GPIO34)

DHT dht(DHTPIN, DHTTYPE);
```

```
void setup()
{
  Serial.begin(115200);
  dht.begin();
}
void loop()
{
  // Read Temperature & Humidity
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

  // Read LDR value
  int lightValue = analogRead(LDR_PIN);

  //Check if DHT failed
  if (isnan(temperature) || isnan(humidity))
  {
    Serial.println("Failed to read from DHT sensor!");
  }
  else
  {
    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.print(" °C | Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");
  }

  // Print Light Data
```

```

Serial.print("LDR Value: ");

Serial.print(lightValue);

if (lightValue < 1000)
{
    Serial.println(" → DARK");
}

else
{
    Serial.println(" → BRIGHT");
}

Serial.println("-----");

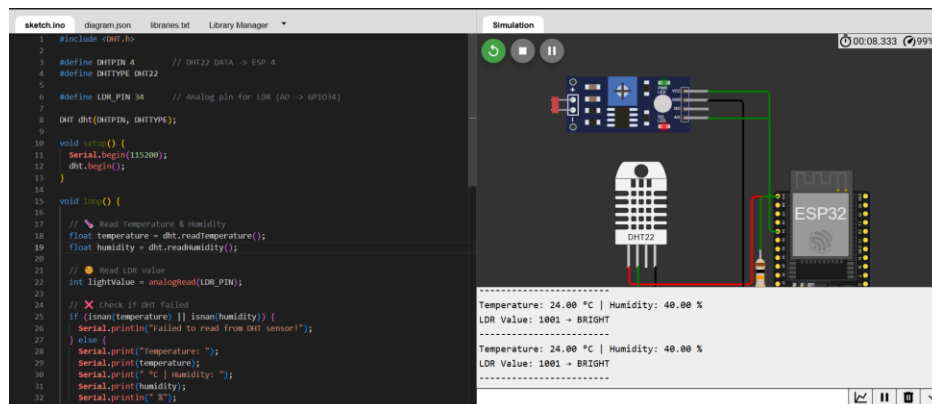
delay(2000);
}

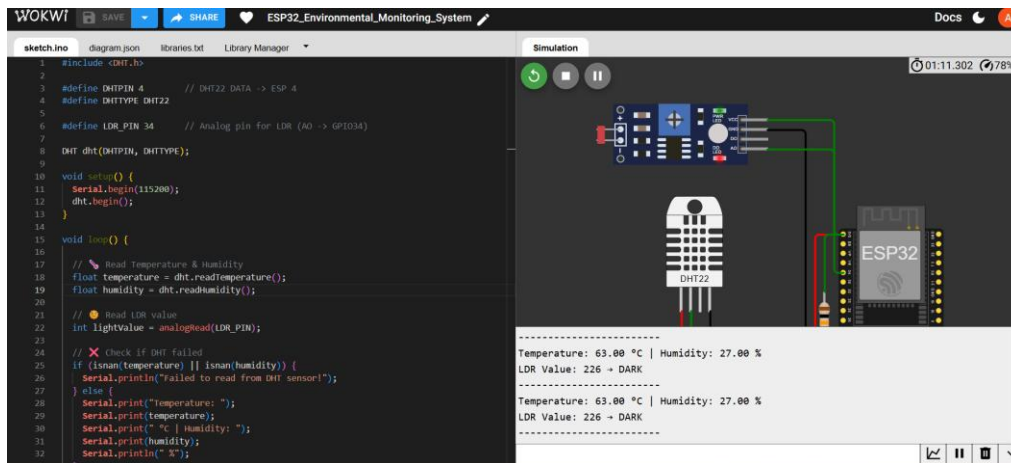
```

8. Output

The serial monitor displays:

- Temperature in °C
- Humidity in %
- LDR value indicating light intensity
- Light condition (Bright/Dark)

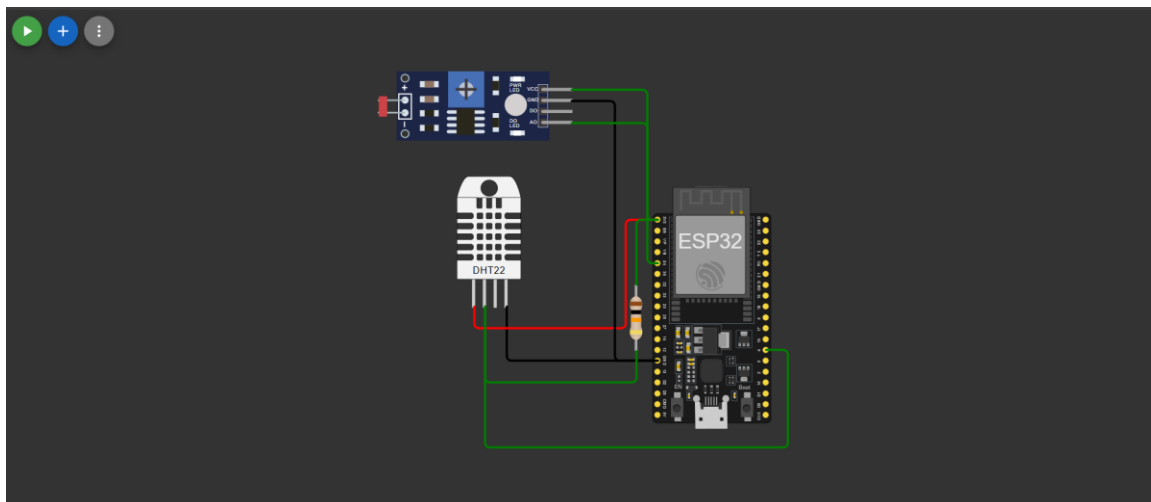




9. Simulation Platform

The serial monitor displays:

- Temperature in °C
- Humidity in %
- LDR value - indicating light intensity
- Light condition (Bright/Dark)



10. Applications

- Smart home automation systems
- Weather monitoring systems
- Agricultural monitoring

- Energy-efficient lighting systems
- IoT-based environmental sensing

11. Conclusion

This project successfully demonstrates how an ESP32 can be used to interface multiple sensors for environmental monitoring. It provides a simple yet effective way to understand sensor integration, analog and digital data processing, and real-time monitoring. The use of simulation tools like Wokwi makes it accessible even without hardware.

12. Wokwi Link

<https://wokwi.com/projects/460563737605955585>

13. Future Scope

- Integration with IoT platforms (ThingSpeak, Blynk)
- Mobile app monitoring
- Data logging and cloud storage
- Adding more sensors like gas or motion sensors